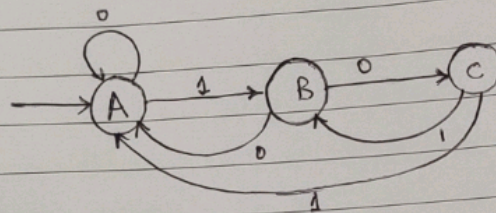


CO1 - Computational Problems

Q1. How can every non-deterministic finite automaton (NFA) be converted into an equivalent deterministic finite Automaton (DFA) and what are the challenges involved? Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language L .



• Introduction :

Every non-deterministic Finite Automaton (NFA) can be converted into an equivalent deterministic Finite Automaton (DFA) using the subset Construction Algorithm (or powerset construction)

- Given NFA

States :

$$Q = \{A, B, C\}$$

Alphabet :

$$\Sigma = \{0, 1\}$$

Initial state : A

Final state : C

From the given NFA :

$$A \xrightarrow{0} A$$

$$A \xrightarrow{1} B$$

$$B \xrightarrow{0} A, C$$

$$C \rightarrow A, B$$

each DFA state represents a set of NFA states.

$$q_0 = A$$

$$q_1 = B$$

$$q_2 = A, C$$

$$q_3 = A, B$$

$$q_d = \emptyset$$

Since q_0 is the final state of the NFA, any DFA state containing q_0 is a final state.
Therefore,

$$F_{DFA} = \{q_0, q_2\}$$

Transition Calculation -

From A ,

$$\text{For input } 0: \delta(\{A\}, 0) = \{A\}$$

$$\text{For input } 1: \delta(\{A\}, 1) = \{B\}$$

From B ,

$$\text{For input } 0: \delta(\{B\}, 0) = \{A, C\}$$

$$\text{For input } 1: \delta(\{B\}, 1) = \emptyset$$

From A, C ,

$$\text{For input } 0: \delta(\{A, C\}, 0) = \{A\}$$

$$\text{For input } 1: \delta(\{A, C\}, 1) = \{A, B\}$$

From A, B ,

$$\text{For input } 0: \delta(\{A, B\}, 0) = \{A, C\}$$

$$\text{For input } 1: \delta(\{A, B\}, 1) = \{B\}$$

From $\emptyset$

There are no possible transitions, so:

$$\delta(\emptyset, 0) = \emptyset$$

$$\delta(\emptyset, 1) = \emptyset$$

- DEA Transition

~~(b) [REDACTED]~~

- Conclusion :

Conclusion :
Using the subset construction method, the given NFA is converted into an equivalent DFA. The DFA has the same accepted language L , while providing exactly one transition for each input symbol from every state.

Q.2. How is mathematical induction used to prove properties of strings and languages in automata theory? Apply the of mathematical induction and prove that for all positive integers n .

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

- Introduction

Mathematical induction is a method used to prove that a statement is true for every positive integer n .

Take : $n = 1$

LHS :

$$\sum_{i=1}^1 i^2 = 1^2 = 1$$

RHS:

$$\frac{1(1+1)(2(1)+1)}{6} = \frac{1 \times 2 \times 3}{6} = 1$$

$$\therefore LHS = RHS$$

Assume that the statement is true for $n=k$.
Therefore,

$$\sum_{i=1}^k i^2 = \frac{k(k+1)(2k+1)}{6}$$

This is called Inductive Hypothesis.

We need to prove that the statement is true for $n = k+1$

$$\sum_{i=1}^{k+1} i^2 = 1^2 + 2^2 + \dots + k^2 + (k+1)^2$$

Using inductive hypothesis:

$$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$= (k+1) \left[\frac{k(2k+1)}{6} + (k+1) \right]$$

$$= \frac{(k+1) [k(2k+1) + 6(k+1)]}{6}$$

$$= \frac{(k+1)(2k^2 + 7k + 6)}{2}$$

This does not match the required formula,

$$\frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

$$= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6}$$

$$= \frac{(k+1)[k(2k+1)+6(k+1)]}{6}$$

$2k^2 + 7k + 6$ is inconsistent

$$\therefore k(2k+1)+6(k+1) = 2k^2 + 7k + 6$$

which factors as $(2k+3)(k+2)$

$$\Rightarrow \frac{(k+1)(k+2)(2k+3)}{6}$$

Now, Since

$$(k+1)+1 = k+2$$

and

$$2(k+1)+1 = 2k+3$$

We get: $\frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}$

Thus the statement holds for $k+1$

(to)

Conclusion:

~~Step 2~~

The statement is true for $n=1$, and if it is true for $n=k$; it is also true for $n=k+1$.

Therefore by the Principle of Mathematical Induction,

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

is true for all positive integers n .

Automaton Simulator: DFA NFA PDA Examples

Test / Debug:

Bulk Testing

Accept (one per line):

```
10
010
0010
1010
01010
```

Reject (one per line):

```
0
1
00
11
101
100
1011
```

```
graph LR
    start((start)) -- ε --> A((A))
    A -- 0 --> A
    A -- 0 --> B((B))
    A -- 1 --> C((C))
    B -- 0 --> A
    B -- 1 --> C
    C -- 1 --> A
    style start fill:#fff,stroke:#000
    style A fill:#fff,stroke:#000
    style B fill:#fff,stroke:#000
    style C fill:#fff,stroke:#000,stroke-width:2px
```

Test Results:

```
Accept: 10 -- Pass
Accept: 010 -- Pass
Accept: 0010 -- Pass
Accept: 1010 -- Pass
Accept: 01010 -- Pass
Reject: 0 -- Pass
Reject: 1 -- Pass
Reject: 00 -- Pass
Reject: 11 -- Pass
Reject: 101 -- Pass
Reject: 100 -- Pass
Reject: 1011 -- Pass
```

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